MISSION EDGESIGHT

TACTICAL EDGE AI-POWERED DECISION INTELLIGENCE FOR TACTICAL-SCALE UNITS

Powered by CALIBRE and Spectro Cloud

Mission EdgeSight is a modular tactical edge AI solution that transforms distributed observations into actionable mission intelligence at the point of need. Developed by CALIBRE and powered by Spectro Cloud's cloud-native software platform, it collects and processes data locally, applies AI-assisted analysis, and delivers operator-ready information through Team Awareness Kit (TAK) compatible workflows to help units accelerate mission understanding in degraded, denied, intermittent, and low-bandwidth (DDIL) environments.

The Tactical Edge Decision Gap

Modern operations generate more observations than tactical-scale units can realistically assess through manual processes alone. Sensors, unmanned systems, cameras, radio frequency (RF) systems, field devices, and external data sources continuously produce information, but information alone does not create understanding. Mission success depends on the ability to determine what changed, where it happened, why it matters, and what action should be taken.

The challenge becomes even greater in DDIL environments where access to centralized analytics may be delayed or unavailable. Raw observations often lack operational context, forcing operators to manually correlate information across systems while making time-sensitive decisions.

Mission EdgeSight was developed to address this tactical edge decision gap by empowering tactical-scale units to sense, process, correlate, assess, and inform closer to the point of action.

The Mission EdgeSight Solution

Mission EdgeSight is built around a simple operational principle: ***the tactical edge should generate mission intelligence at the point of need.***

The solution connects forward sensing, persistent mission state, AI-assisted analysis, operational context, and TAK-integrated delivery within a modular edge architecture designed for disconnected and contested environments. Mission EdgeSight follows a repeatable operational workflow:

- **Sense and Persist:** Observations are collected, preserved, and normalized close to their source. Local buffering and store-and-forward mechanisms help maintain continuity when communications are disrupted.
- **Enrich and Assess:** AI-assisted services correlate observations with mission geography, routes, areas of interest, and operational context. The system helps identify mission-relevant activity while providing evidence and confidence information to support informed decisions.
- **Deliver and Sustain:** Operator-ready mission products are delivered through TAK-compatible workflows, dashboards, and mission applications. The architecture is designed to support repeatable deployment, ongoing sustainment, and adaptation as mission requirements evolve.

Mission EdgeSight combines CALIBRE's mission engineering expertise and edge AI capabilities with Spectro Cloud's cloud-native platform to rapidly develop, deploy, operate, and evolve mission capabilities at the tactical edge.

Spectro Cloud provides the secure, repeatable foundation behind the solution, consistently deploying and managing infrastructure, applications, AI services, and mission technologies across distributed environments. By simplifying deployment, securing and governing the technology stack, and automating lifecycle operations, it reduces operational burden while enabling fleet-wide consistency at scale. Its modular, cloud-native architecture brings greater flexibility, technology choice, and adaptability to the tactical edge, allowing new sensors, AI models, hardware platforms, and mission applications to be integrated as mission requirements evolve. The result is a resilient framework that helps organizations accelerate mission capability delivery and sustain operations across DDIL environments.

Core Capabilities and Operational Effects

Core Capability	Operational Effect
Forward Sense-to-Decision Processing	Processes observations closer to the point of collection, reducing reliance on centralized systems and accelerating mission understanding
Mission Context & AOI Decision Intelligence	Correlates observations with operational geography, routes, areas of interest, and mission priorities to identify what matters most

Core Capability	Operational Effect
Persistent Mission State	Preserves observations, history, and context to support consistent assessment and informed decision-making
TAK-Native Delivery	Delivers operator-ready information directly into familiar mission workflows and common operating pictures
Human-Centered Decision Support	Provides evidence, confidence, and recommendations while preserving human decision authority
Repeatable Mission Deployment & Sustainment	Combines reusable CALIBRE Mission Packs with Spectro Cloud's lifecycle management foundation to support repeatable deployment, governance, sustainment, and scalability across distributed edge environments

Representative Mission Applications

Mission EdgeSight can be applied across a variety of operational environments where timely decisions depend on local understanding and resilient operations. The demonstrated capability helps units generate mission-relevant insights independently in DDIL environments while synchronizing information and leveraging reach-back resources when connectivity is available.

- **Route Awareness and Contested Logistics**: Support movement planning and route assessment by correlating observations with planned routes, infrastructure, and mission-defined areas of interest.
- **Special Operations and Expeditionary Missions:** Enable local processing, mission awareness, and decision support for distributed units operating with limited infrastructure and intermittent connectivity.
- **Force Protection and Critical Infrastructure Security:** Help identify activity that may affect personnel, facilities, infrastructure, and other protected assets through mission-context analysis and geographic correlation.
- **Homeland Security and First Responder Operations:** Support disaster response, border security, emergency management, search and rescue, and public-safety missions where reliable connectivity cannot be guaranteed.
- **AI at the Point of Need:** Bring AI-assisted analysis closer to the mission, reducing latency while generating actionable insights where observations occur.
- **Information Dominance in DDIL Environments:** Maintain mission awareness and decision support during communications disruptions by preserving processing, context, and operator workflows at the tactical edge.

Where This Stands Today

Mission EdgeSight has demonstrated a complete, integrated sense-to-decision workflow, from field sensor collection and local processing through AI-assisted assessment, mission-context generation, TAK publication, and operator presentation in integrated testing. CALIBRE self-assesses the integrated capability at Technology Readiness Level (TRL) 6, with maturity varying by subsystem: core capabilities, including sensing, buffering, mission-state processing, and TAK publication, are more mature, while advanced multi-sensor fusion and adaptive recommendations remain prototype or roadmap items.

CALIBRE recommends a mission-focused pilot, aligned to the organization's mission requirements, sensors, operational environment, and communications conditions, as the appropriate next step before operational use. Such an evaluation would assess performance, mission utility, cybersecurity considerations, and operator effectiveness in the intended environment.

Next Step

CALIBRE will partner with your organization to evaluate Mission EdgeSight against mission requirements, operating conditions, and sensor ecosystems through a mission-focused pilot. The evaluation will demonstrate how tactical edge sensing, local AI-assisted analysis, mission context, and TAK-enabled decision support can accelerate mission understanding, strengthen situational awareness, and improve decision-making in DDIL environments.

Contact CALIBRE to discuss a mission-focused pilot of Mission EdgeSight and explore how the capability may support your operational objectives.

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